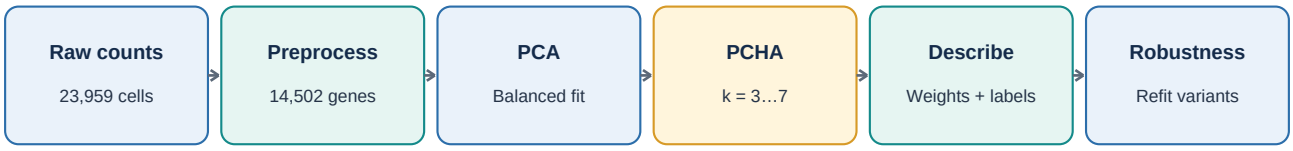
MOUSE SCC ARCHETYPE PIPELINE

From intuition to mathematical understanding to scientific defence

A detailed learning guide to the analysis of 23,959 mouse squamous cell carcinoma cells from GSE288137: preprocessing, PCA, PCHA archetypes, EMT-state mapping, model selection, confounding and robustness.



Fit a geometric model first; interpret biological labels afterwards.

The central question

Can the variation among mouse SCC cells be described by a small number of extreme expression profiles, and where do the paper's EMT states lie relative to those profiles?

How to use this guide

- **First pass:** read the intuition and pipeline overview without worrying about equations.
- **Second pass:** study the mathematical sections and explain each equation in plain language.
- **Before a meeting:** practise the defence questions and verify the implementation-specific details against the actual scripts and outputs.

Scope note: this guide explains the pipeline specification provided in the conversation. It does not report fitted results or claim a winning value of k. Where exact implementation details are not specified, it flags what must be checked in source code.

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One rule to keep in mind

A pipeline can produce a mathematically valid geometric fit without proving that the geometry is driven by EMT. The job of interpretation and robustness is to distinguish those claims.

1. The big picture: what question is being asked?

The dataset contains 23,959 mouse squamous cell carcinoma (SCC) cells. Each cell is represented by an expression profile measured across thousands of genes. The pipeline fits a low-dimensional geometric model to those profiles, then asks how independently defined EMT states relate to the fitted geometry.

Think of each cell as a point described by thousands of measurements. The analysis first compresses those measurements into a manageable coordinate system, then finds a small set of extreme profiles that can be mixed to approximate the cells.

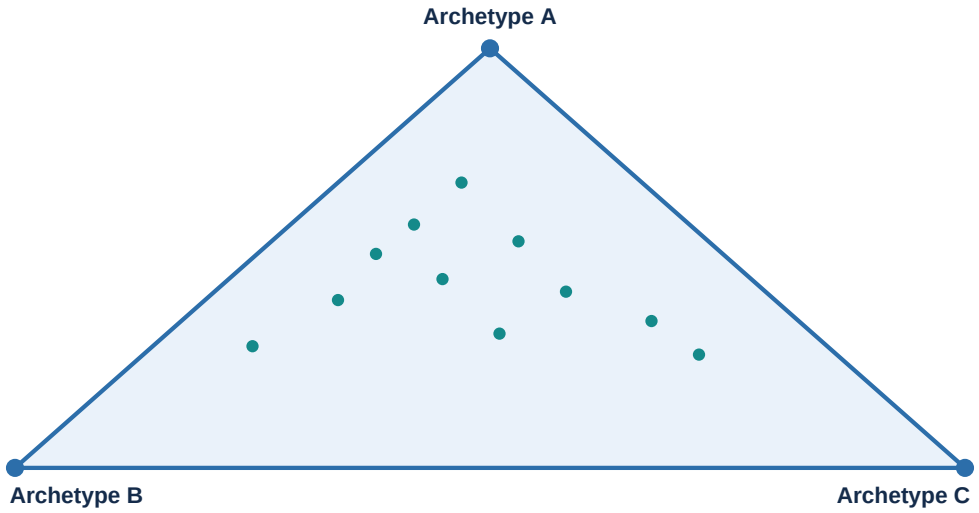


Illustration only — not fitted results from GSE288137.

Figure 1. A three-archetype simplex. Each point is a cell and each corner is a fitted extreme profile. Points near a corner have a more dominant weight for that archetype; points in the interior have more mixed weights. The figure is explanatory, not a result from this dataset.

The project has two linked but distinct questions

- **Geometry question:** does a low-dimensional simplex describe the expression structure in a way that is useful relative to the chosen null and model assumptions?
- **Biology question:** do the EMT states from the reference paper map to particular vertices, edges or interior regions in a reproducible way?

Why separate these questions?

If EMT labels determine the fitted vertices, then finding EMT labels near those vertices is partly circular. The pipeline instead fits the geometry without using the EMT labels to choose it, then investigates the label–geometry relationship.

The pipeline at a glance

Step	Input / action	Output / purpose
1. Build AnnData	Counts → CP10k → log1p; filter genes	Expression matrix with 14,502 genes
2. PCA	Balanced fit subset; exact PCA; project all cells	Cell coordinates and PC diagnostics
3. PCHA fits	Fit k = 3…7; ESV and permutation test	Candidate simplices and fit evidence
4. Describe simplex	Weights, labels, tumours, genes, enrichment	Biological characterisation
5. Robustness	LOTO, scRNA-only, nophase, seed	Sensitivity of conclusions

2. Single-cell RNA-seq and the input matrix

Single-cell RNA sequencing (scRNA-seq) measures RNA molecules captured from individual cells. After sequencing and processing, the data are often represented as a gene-by-cell count matrix. A count is a measurement of captured molecules or reads attributed to a gene; it is not a direct, noise-free measurement of the cell's true biological activity.

A toy matrix

Gene	Cell 1	Cell 2	Cell 3
Gene A	120	15	70
Gene B	5	200	30
Gene C	60	45	150
Gene D	0	12	3

Rows are genes; columns are cells. Your real matrix is much larger. In the supplied pipeline description, the final filtered matrix has 14,502 genes across 23,959 cells.

Why is the data noisy?

- **Sampling:** only a fraction of RNA molecules in a cell may be captured and sequenced.
- **Library size:** cells can have different total numbers of captured counts.
- **Sparsity:** many gene–cell entries are zero, which can mean truly low expression or failure to detect a transcript.
- **Biological heterogeneity:** cells genuinely differ in state, differentiation, cell cycle, stress response and other programmes.
- **Technical and experimental structure:** tumour of origin, library preparation, assay type and processing can affect the measured expression profiles.

A key distinction

Observed expression contains both biological variation and measurement-related variation. Preprocessing can improve comparability, but it cannot guarantee that the remaining variation is the biological process you care about.

What is SCC?

Squamous cell carcinoma is a cancer arising from squamous epithelial cells. In this project, the biological focus is how cells within mouse SCC relate to EMT-associated states. EMT is not assumed to be a simple on/off switch; hybrid or partial EMT states may retain some epithelial features while acquiring mesenchymal features.

3. Step 1 — preprocessing the expression matrix

Script: **01_build_anndata.py** • stated runtime: about 5 minutes

The script transforms raw counts to log1p CP10k, retains genes detected in at least 1% of cells, and removes mitochondrial and ribosomal genes. The stated result is 14,502 retained genes.

3.1 Library-size normalisation: CP10k

Cells can have different total counts because of capture and sequencing depth. Counts-per-10,000 (CP10k) rescales each cell's counts so that its total across genes is 10,000 before the log transform.

$$\mathrm{CP10k}(g,c) = \mathrm{count}(g,c) / \mathrm{total_counts}(c) \times 10{,}000$$

Here, count(g,c) is the raw count for gene g in cell c, and total_counts(c) is the sum of counts across the genes used in the normalisation denominator.

Cell	Gene A	Gene B	Total counts
Cell 1 raw	100	50	1,000 (illustrative)
Cell 2 raw	1,000	500	10,000 (illustrative)

If the relative composition is the same but one cell has ten times the library size, normalisation brings the corresponding relative expression values onto a comparable scale. It does not make the cells biologically identical.

3.2 Log1p transformation

After normalisation, the pipeline uses log1p: log(1 + x). The +1 makes zero values well-defined, and the logarithm compresses large values. This reduces the leverage of very high expression values and often makes variation easier to model.

Example: log1p(0) = 0; log1p(9) ≈ 2.30; log1p(99) ≈ 4.61. A tenfold increase does not remain a tenfold increase on the transformed scale.

3.3 Detection filter: genes found in at least 1% of cells

One percent of 23,959 cells is approximately 240 cells. Genes detected in fewer cells are excluded under this rule. This can reduce the influence of extremely sparse features, but it may also remove a genuine marker expressed only in a rare population.

3.4 Removing mitochondrial and ribosomal genes

Mitochondrial genes can reflect metabolic state, stress or cell quality. Ribosomal genes can reflect translation-related programmes and other sources of variation. Removing them is intended to prevent these gene families from dominating the fit, but it can also remove biologically meaningful signal. This is a deliberate scope choice, not a universal rule.

What preprocessing does not do

CP10k + log1p + filtering do not perform batch correction, remove tumour identity, eliminate assay differences or prove that the retained genes are EMT-specific. Those questions are handled later through diagnostics and robustness analyses.

Author-level explanation

> *“We normalise counts per cell to reduce library-size differences, apply a log transform to compress the dynamic range, and filter rare and mitochondrial/ribosomal features. The result is a more tractable expression matrix, while recognising that technical and biological confounding may remain.”*

4. Step 2 — PCA: compressing expression into coordinates

Script: **02_pca.py --geneset all** • stated runtime: about 2 minutes

PCA stands for Principal Component Analysis. It replaces thousands of correlated gene measurements with a smaller set of orthogonal axes that capture the largest directions of variation in the data.

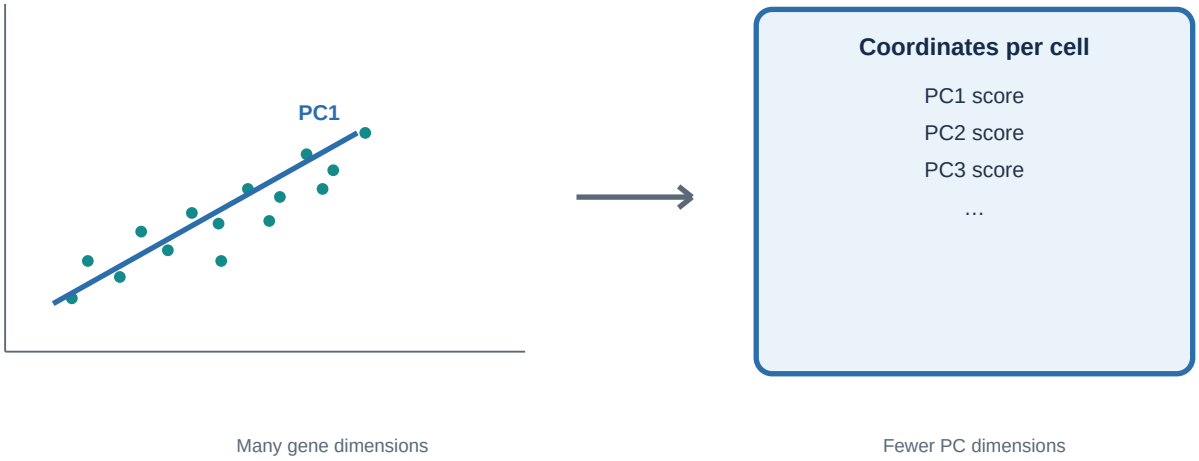


Figure 2. PCA converts each cell's high-dimensional gene-expression profile into scores on a smaller number of principal components. Each PC is a weighted combination of genes, not a single gene.

4.1 What is a principal component?

A principal component (PC) is a direction in expression space. PC1 is the direction that captures the most variance under the PCA objective. PC2 captures the largest remaining variance subject to being orthogonal to PC1, and so on. Each cell receives a score on every retained PC.

For a centred matrix X with cells as rows and genes as columns, the first PC direction can be written as:

$$v_1 = \arg\max_{\|v\|=1} \|Xv\|^2$$

The vector v contains gene weights (loadings), while Xv gives the PC scores across cells. PCA is unsupervised: it does not know which cells are epithelial, hybrid or late-EMT.

4.2 Why use PCA before archetype analysis?

- The original expression space has 14,502 dimensions.
- Many genes are correlated, so the data may be described by fewer major variation directions.
- Fitting a simplex in a manageable number of dimensions is computationally easier.
- The selected PC space defines which expression variation the archetype model can capture.

The danger of variance-first analysis

PC1 can be dominated by tumour, assay type, cell cycle or other effects. High variance is not the same as high biological relevance. PC diagnostics are therefore part of the scientific analysis, not just a technical plot.

4.3 Why tumour-balanced fit cells?

The pipeline caps the PCA fitting subset at 800 cells per tumour. With 12 tumours, the maximum balanced fitting sample is 9,600 cells. All 23,959 cells are then projected into the fitted PCA space.

This reduces the chance that the largest tumours determine the PC directions simply because they contribute more cells. It does not guarantee equal biological diversity across tumours or eliminate confounding. Sampling fewer cells from a tumour can also omit rare states, so the cap is a trade-off.

4.4 Exact PCA through the Gram matrix

Let X be the centred matrix with n cells and p genes. The Gram matrix is G = XX■, an n × n matrix of cell-to-cell inner products. When p is large relative to n, decomposing the cell-by-cell matrix can be more efficient than directly decomposing a gene-by-gene covariance matrix.

The corresponding decompositions share the nonzero eigenvalues; PC scores can be recovered with the appropriate scaling. The pipeline reports score correlation 1.0000000 against full SVD on this dataset. This supports numerical agreement, allowing for sign conventions and potential rotations in nearly degenerate subspaces.

Three separate justifications

PCA: dimensionality reduction. **Tumour balancing:** reduce overrepresentation during fitting. **Gram PCA:** compute the same PCA efficiently enough for repeated refits. Do not treat these as interchangeable reasons.

5. Archetype analysis: the geometric core

The PCA step provides coordinates. PCHA then estimates a small set of extreme profiles—archetypes—that can represent the observed cells as mixtures.

5.1 What is an archetype?

An archetype is a mathematically estimated extreme expression profile. It is not necessarily an actual observed cell, and it is not automatically a known cell type or biological mechanism.

5.2 Convex combinations

A convex combination is a weighted average in which all weights are non-negative and sum to one. In a simplified archetype formulation, archetype j is represented as a convex combination of observed cells:

$$z_j = \sum_i \alpha_{ij} x_i, \alpha_{ij} \geq 0, \sum_i \alpha_{ij} = 1$$

Each observed cell is then approximated as a convex combination of the archetypes:

$$x_i \approx \sum_j \beta_{ij} z_j, \beta_{ij} \geq 0, \sum_j \beta_{ij} = 1$$

The β weights describe where a cell lies relative to the archetypes. These equations explain the convex-mixture intuition; the precise constraints and objective should be checked in the PCHA implementation used by the project.

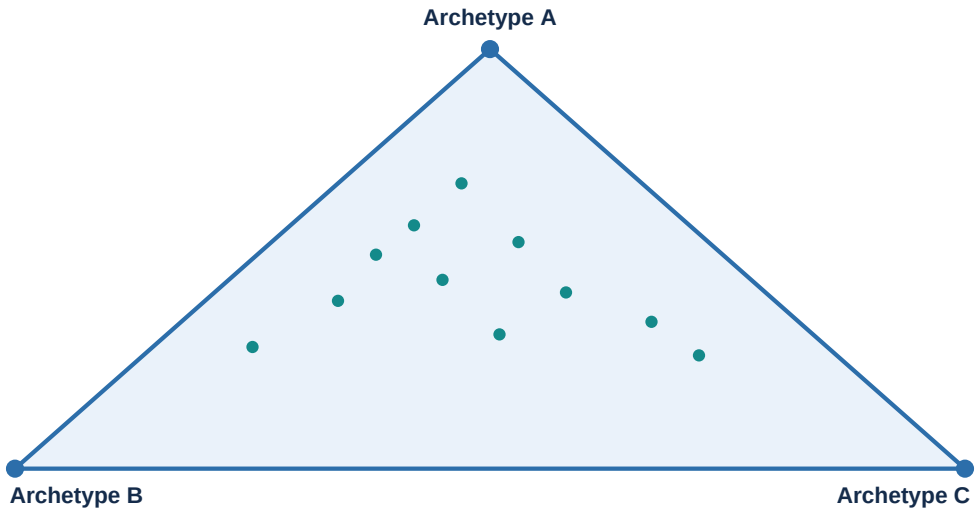


Illustration only — not fitted results from GSE288137.

Figure 3. With three archetypes, the simplex is a triangle. With four, it is a tetrahedron in three dimensions. In general, k vertices define a (k–1)-dimensional simplex.

5.3 Interpreting a cell's position

- **Near a vertex:** one archetype weight is dominant.
- **Near an edge:** two archetypes may contribute strongly.
- **In the interior:** the representation is more mixed across archetypes.

These are geometric descriptions. A cell near an edge is not automatically a transitional cell, and a cell near a vertex is not automatically a stable cell type. Biological meaning comes from state labels, gene programmes and robustness evidence.

5.4 What is PCHA?

PCHA (Principal Convex Hull Analysis) is an archetype-fitting method. The pipeline uses the implementation in src/archetypes.py, fits on the first k–1 PCs, sets delta = 0, runs 15 starts and retains the largest-volume solution. The candidate values are k = 3, 4, 5, 6 and 7.

Because the optimisation is non-convex, different initialisations may converge to different local solutions. Multiple starts reduce dependence on a single initialisation but do not guarantee the global optimum.

5.5 What does delta = 0 mean?

In the common PCHA objective, delta controls a volume-related regularisation term. Setting delta to zero removes that penalty contribution; it does not mean that the fitted simplex has zero volume. Confirm the exact objective and parameter convention in the project's implementation before giving a formal derivation.

5.6 Why keep the largest-volume solution?

The largest-volume solution may capture a more extreme simplex under the specified fitting objective. But largest volume does not automatically mean lowest reconstruction error, best biological interpretation or greatest reproducibility. That is why model diagnostics and refits remain necessary.

5.7 Archetypes versus clustering

Clustering	Archetype analysis
Groups cells into discrete groups.	Fits extreme profiles and represents cells as mixtures.
A hard assignment is common, though soft clustering exists.	Weights are continuous and sum to one in the convex representation.
Focuses on groups of similar observations.	Focuses on extremes and relationships among them.
Does not inherently model a continuum between cluster centres.	Can represent intermediate locations, but does not prove a biological trajectory.

Do not overinterpret geometry

Archetype weights describe a fitted mathematical representation in the selected PC space. They are not probabilities of cell identity, measures of causal influence or evidence that cells physically travel along a trajectory.

6. Choosing the number of archetypes

The pipeline fits k = 3 through k = 7. It uses an ESV curve and a permutation test as complementary pieces of evidence. The pipeline description does not specify which k ultimately wins; that decision must be grounded in the actual outputs and preregistered criteria.

6.1 ESV curve and the elbow heuristic

The ESV curve tracks the behaviour of a fit-related quantity as the number of archetypes increases. An elbow is a point after which additional archetypes provide diminishing returns. It is a heuristic, not a formal significance test, and the exact definition of ESV should be verified in the code.

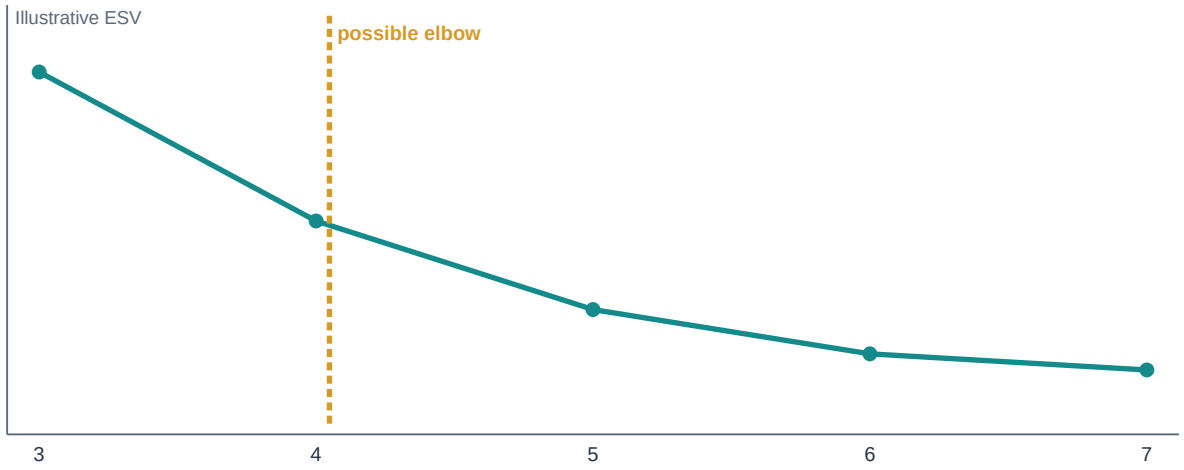


Figure 4. Conceptual illustration of a diminishing-returns curve. Do not read the displayed values as the actual ESV results.

Why can the elbow be ambiguous?

- The curve may decline smoothly without a clear bend.
- A higher k can improve the numerical fit without adding a meaningful biological dimension.
- A visually appealing elbow can be selected differently by different analysts.
- The curve alone does not show whether vertices are stable across tumours or assays.

6.2 The permutation test

The pipeline independently shuffles each of the first k–1 PC coordinates, refits PCHA using five starts, and compares the observed fit statistic with the shuffled null distribution. This destroys the original joint organisation of the coordinates while retaining the marginal values within each PC.

The estimated permutation p-value is:

$$p = (1 + b) / (1 + B)$$

Here B = 200 shuffles and b is the number of shuffled fits at least as good as the observed fit under the implemented statistic. If b = 0, the smallest possible estimate is $1/201 \approx 0.00498$, not zero.

6.3 What the null does—and does not—test

A small p-value says that the observed statistic is unusual relative to this specific independently shuffled null. It does not prove that the data have a biological Pareto front or that the geometry is driven by EMT. Tumour and assay structure can also produce non-random structure that the shuffle destroys.

Interpretation rule

Statistical significance is always relative to a null model. A p-value cannot answer a broader question than the null and statistic were designed to test.

6.4 How to select k responsibly

- Examine the ESV curve and whether extra vertices yield useful improvement.
- Interpret the permutation test relative to its precise null and statistic.
- Check whether vertex profiles and cell-state associations are stable under refitting.
- Assess tumour and assay dependence.
- Use preregistered criteria where possible; do not select k only because it produces the preferred EMT narrative.

7. Step 4 — describing the fitted simplex

Script: **04_describe_simplex.py --geneset all --k K** • stated runtime: about 2 minutes

This step calculates weights for every cell, creates label tables, checks tumour composition, performs enrichment analysis, identifies marker genes and produces figures.

7.1 Cell weights

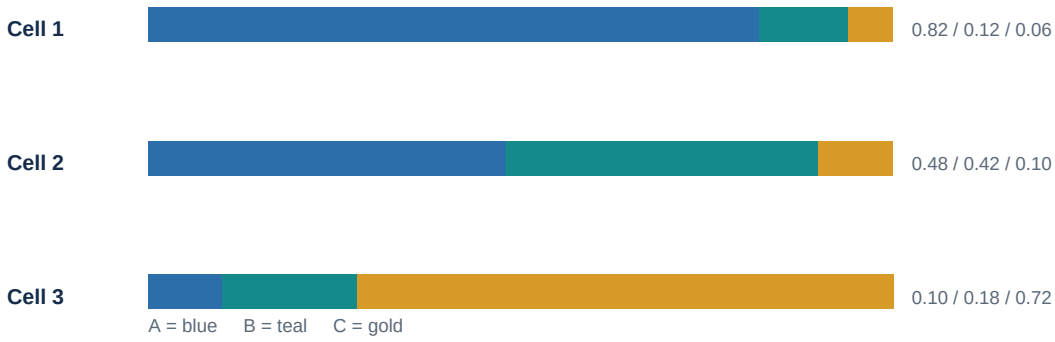


Figure 5. Example weight vectors for three illustrative cells. Each vector sums to one; colours denote archetypes A, B and C. Values are invented to demonstrate interpretation.

If a cell has weights 0.82, 0.12 and 0.06, its representation is dominated by archetype A. If it has 0.48, 0.42 and 0.10, it is more mixed. These weights are coordinates in the model, not calibrated probabilities that the cell belongs to biological classes.

7.2 Specialist rule

The pipeline defines a specialist cell when the largest weight exceeds the second-largest weight by at least 0.35:

$$w(1) - w(2) \geq 0.35$$

Weights	Largest minus second	Specialist?	Reason
0.80, 0.15, 0.05	0.65	Yes	Clear dominance
0.48, 0.42, 0.10	0.06	No	Top two are similar
0.50, 0.15, 0.35	0.15	No	Gap below threshold

The rule helps distinguish cells with a dominant archetype from cells whose weights are close. The threshold is inherited from the Pal work; it is not a universal biological boundary. If conclusions depend on 0.35, test reasonable alternative thresholds.

7.3 EMT-state label tables

The paper's state labels are compared with weights and specialist assignments. You can ask whether a state is overrepresented near a vertex, spread across the simplex, or associated with more than one archetype. These are associations; they do not establish causality or a temporal trajectory.

7.4 Tumour breakdown

The same vertex may look like an EMT programme but actually be dominated by a particular tumour. Because four tumours contribute 75% of late-EMT cells, checking tumour composition is especially important. A convincing biological pattern should not be an artefact of one sample's overrepresentation.

7.5 Marker genes and enrichment

Marker genes help describe the expression programmes associated with a vertex or group of cells. Enrichment analysis asks whether genes or labels are overrepresented relative to a specified background. You must inspect the actual test, background set, multiple-testing correction and definition of the tested gene list before interpreting a reported enrichment result.

Potential circularity in marker interpretation

The archetypes are fitted from the same expression data used to describe them. Marker genes therefore help characterise the fit but are not fully independent validation. Independent signatures, held-out samples or external datasets provide stronger validation.

8. EMT biology and label-independent mapping

EMT is a biological programme in which epithelial cells acquire mesenchymal-associated characteristics. In cancer, EMT-associated changes can include altered adhesion, polarity, cytoskeletal organisation and motility. Many tumours contain partial or hybrid states rather than a clean binary switch.

A simplified conceptual continuum

Epithelial-like Adhesion and polarity-associated features	**Hybrid / partial EMT** Combination of epithelial and mesenchymal features	**Mesenchymal-like** Mesenchymal-associated programmes

This is a simplified illustration, not a claim that all cells follow one linear path. EMT state labels may represent operational definitions based on the reference paper, and the mapping must respect how those labels were actually assigned.

8.1 Why fit without EMT labels?

The model is fitted without using EMT labels to determine the vertices. The labels are used afterwards to examine where the states sit. This reduces direct circularity: the fit is not explicitly instructed to create an epithelial corner, a hybrid corner and a late-EMT corner.

However, label-independent fitting is not the same as independent validation. The same cells and expression measurements may still be used for both fitting and interpretation, and model selection can introduce bias. The strength comes from keeping the model fit separate from the biological interpretation and testing the interpretation for robustness.

8.2 Possible outcomes

Observed pattern	Possible interpretation	What to check
Epithelial state favours one vertex	A vertex may capture an epithelial-associated programme.	Marker genes, tumour distribution, assay composition.
Late-EMT state favours another vertex	A vertex may capture a late-EMT-associated programme.	Consistency across tumours and scRNA-only refit.
Hybrid cells occupy edges/interior	Could be consistent with mixed expression profiles.	Confirm labels and weights; do not infer trajectory automatically.
Each state spreads across vertices	EMT labels may not align with the fitted extremes, or other variation dominates.	Assay/tumour effects, k choice, alternative gene set.
Vertices correspond mainly to tumours	Geometry may reflect sample-specific structure.	LOTO, assay diagnostics and external data.

8.3 What the analysis cannot establish by itself

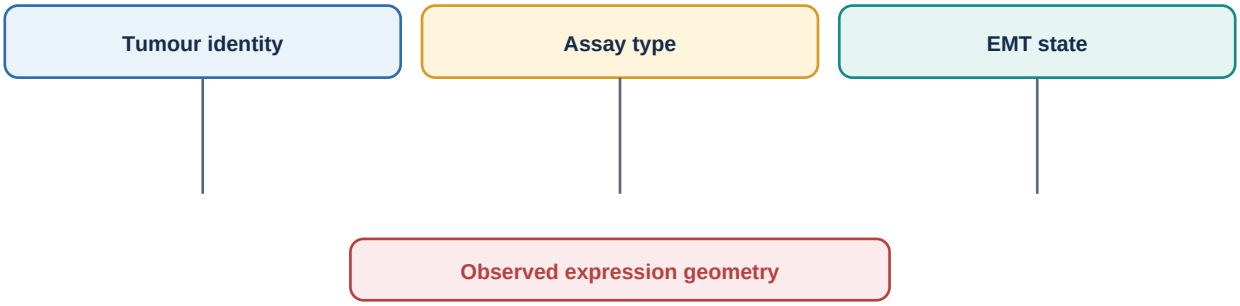
- That cells physically move along the simplex during EMT.
- That an archetype causes EMT or controls cell behaviour.
- That the weights are transition probabilities or pseudotime.
- That the fitted vertices correspond to stable cell types.
- That an association with late EMT is independent of all tumour and assay effects.

A careful biological claim

“The fitted archetype weights show a reproducible association between the reference EMT-state labels and regions of the simplex.” This is appropriately narrower than claiming that the model proves a trajectory or causal mechanism.

9. Confounding: tumour identity, assay type and EMT

The pipeline notes that, using all genes, PC1 is explained more by tumour and assay (multiome nuclei versus whole cells) than by EMT state. This is a central limitation because the model will fit the largest variation it sees, not the variation the researcher wishes it would see.



Conceptual: multiple causes can shape the same apparent pattern.

Figure 6. Conceptual illustration: tumour identity, assay type and EMT state can all influence observed expression geometry. The same apparent separation may have multiple explanations.

9.1 What is confounding?

A confounder is a variable associated with both the pattern being studied and an alternative explanation for it. In this setting, tumour identity or assay type may be associated with cell-state composition and expression profiles. If these factors align, the model may interpret them as major axes or vertices.

Example: if a particular tumour contributes many late-EMT cells and was processed using a different assay, then a vertex enriched for those cells could reflect EMT, tumour-specific biology, assay differences or a mixture of all three.

9.2 Why not batch-correct automatically?

Batch correction can reduce technical differences, but it can also remove genuine biological variation when batch and biology are entangled. If a tumour is observed only in one assay, the dataset may not contain enough information to separate tumour biology from assay effect cleanly. The choice to fit without batch correction preserves raw structure for inspection, but it also means confounding must be addressed explicitly.

9.3 Why tumour balancing is not enough

- Balancing limits cell-count domination during PCA fitting.
- It does not equalise biological diversity, assay composition or state distributions.
- It does not remove systematic technical differences.
- It does not guarantee that rare cell states are represented in the fitting subset.

The key question

Could tumour identity or assay type explain the same pattern that we are interpreting as EMT? Every strong biological interpretation should address this alternative.

10. Step 5 — robustness and sensitivity analysis

Script: **05_robustness.py --geneset all --k K --variant loto** • stated runtime: about 3 minutes per refit

Robustness analysis asks whether the main conclusions depend on a particular sample, assay subset, preprocessing choice or random initialisation.

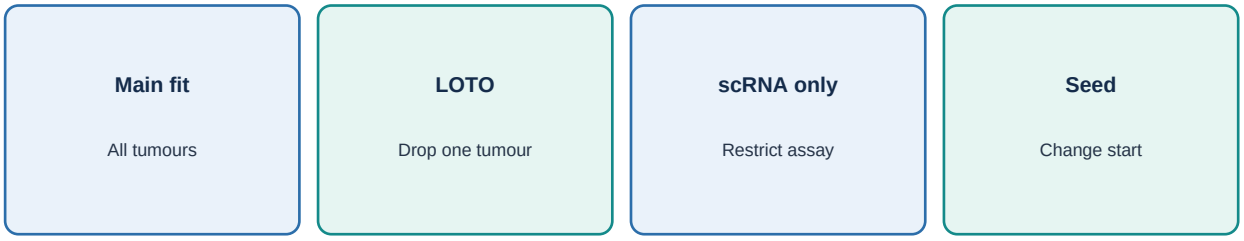


Figure 7. The main fit is compared with several refitting variants. Compare corresponding biological conclusions, not only raw vertex coordinates.

10.1 LOTO: leave one tumour out

The model is refitted repeatedly, excluding one tumour each time. This tests whether a particular tumour is essential to the apparent geometry. If a vertex or EMT association disappears whenever one tumour is removed, the claim is sample-sensitive.

Important: LOTO is a sensitivity analysis, not fully independent validation. The remaining tumours still come from the same study and may share technical effects.

10.2 scRNA-only

This variant restricts the analysis to the scRNA-seq subset according to the pipeline's assay filter. It tests whether the geometry persists without multiome-nuclei data. Confirm exactly which cells are included and whether the PCA is refitted within the subset or uses a projected space; the script determines the precise interpretation.

10.3 nophase

The name suggests an alternative treatment of a phase-related variable or filter, but the pipeline summary does not define the exact operation. Do not guess. Inspect the script to establish what changes, what remains fixed and why that is a relevant sensitivity test.

10.4 seed

This variant examines sensitivity to random choices such as initialisation. Verify whether it changes only the random seed, the sampled fitting cells, the PCHA starts, or multiple components. Different vertices can be permuted or oriented differently across fits, so compare profiles and biological associations rather than relying on vertex labels alone.

10.5 How to compare models across refits

- Match vertices by their expression profiles, weight patterns or another prespecified similarity criterion.
- Account for vertex label permutations; “vertex 1” in one fit need not be “vertex 1” in another.
- Compare EMT-state associations, tumour representation and marker programmes.
- Check whether k remains defensible and whether fit quality degrades substantially.

- Report instability rather than forcing a consistent biological name onto unmatched vertices.

Robustness is not a pass/fail ritual

A robust result is not necessarily true, and an unstable result is not automatically useless. Robustness tests show how dependent the conclusion is on specific assumptions and samples. Interpret each variant in light of the threat it is designed to probe.

11. All genes versus the 176-gene KS panel

The pipeline supports two feature sets. The **all** set uses the 14,502 genes retained after preprocessing. The **ks** set uses the 176 Tan 2014 KS genes found in mouse, described as the panel used for the Pal breast fits.

Feature set	What it asks	Potential strength	Potential limitation
all	What structure appears across the retained expression space?	Broad, data-driven view of variation.	Strong non-EMT variation may dominate.
ks	What structure appears when analysis is restricted to the shared 176-gene panel?	More directly comparable with the earlier breast analysis.	Narrow panel may omit important SCC biology; mapping may lose genes.

A change in gene set changes the geometry

PCA directions depend on which genes are included. Therefore, the same cells can have different PC scores and different fitted archetypes under different gene sets. A fit on 176 genes is not numerically interchangeable with a fit on 14,502 genes.

The panel comparison can test whether a broad pattern is also visible in a predefined feature set, but the KS panel is not an independent validation dataset. It is an alternative feature-selection strategy.

Mouse–human orthology considerations

If the original panel was defined in another species, the mouse gene mapping needs careful treatment. One-to-one orthologues, many-to-one mappings, absent genes and gene-symbol conventions can affect the final feature set. The precise mapping procedure should be documented.

Comparability is not equivalence

Using the same named panel as earlier work helps align feature selection, but differences in species, tissue, assay, preprocessing and sample composition can still change the result.

12. Statistical interpretation and common failure modes

This section separates what the pipeline directly measures from what a researcher may be tempted to infer.

Evidence	Reasonable conclusion	Conclusion not justified by itself
A clear ESV elbow	A candidate k may balance fit and complexity.	This is the uniquely correct number of biological states.
Small permutation p-value	Observed fit statistic is unusual under the specified shuffle null.	The geometry is biological or EMT-driven.
One dominant weight	The cell is represented mainly by one fitted archetype.	The cell has a known type with that probability.
EMT-state enrichment	A state is associated with a vertex/group under the chosen test.	The vertex causes EMT or cells move through it.
Stable LOTO results	The result is not dependent on any single omitted tumour under this procedure.	The model generalises to all SCCs.
Marker genes look plausible	The vertex is described by a coherent expression programme.	The markers independently validate the model.
All genes and KS agree	Some conclusions may be robust to feature selection.	The result is independently replicated.

12.1 Multiple testing

When many genes, labels or pathways are tested, some will appear enriched by chance. A multiple-testing procedure such as Benjamini–Hochberg false-discovery-rate correction controls the expected proportion of false discoveries among rejected hypotheses under its assumptions. Confirm whether and how the actual script applies correction.

12.2 Cells are not independent biological replicates

Thousands of cells from the same tumour do not provide thousands of independent tumour-level replicates. Cells share sample-level context and may be correlated. For claims about generalisation across tumours, the tumour is often the relevant independent unit, and cell-level enrichment should be interpreted with that structure in mind.

12.3 Model selection and post-selection interpretation

Trying multiple values of k, feature sets and variants is useful, but it creates opportunities to favour a model that tells the preferred story. Preregistered predictions help by recording expectations before the fit. They do not eliminate all analytical flexibility; deviations from the plan and exploratory follow-ups should be disclosed.

12.4 Correlation, association and causation

A relationship between EMT labels and archetype weights is associational. A fitted geometry does not establish that one programme causes another, identify the direction of a transition, or show that individual cells physically move through the simplex. Those claims require additional evidence such as time-resolved data, perturbations or independent experiments.

13. Defending the pipeline in a research meeting

Use these answers as starting points. They are deliberately careful: they explain the rationale without claiming more than the method establishes.

Q1. What is the research question?

"We want to characterise the geometric organisation of mouse SCC expression profiles using archetype analysis, then investigate how the reference paper's EMT states map onto that structure. We do not assume in advance that the cells form three discrete EMT classes or a single linear trajectory."

Q2. Why archetypes rather than clustering?

"Clustering groups similar cells, while archetype analysis estimates extreme profiles and represents cells as mixtures. The continuous weights let us investigate whether EMT states are associated with extremes, edges or intermediate regions. The method is complementary to clustering and does not itself prove a trajectory."

Q3. Why PCA first?

"The expression matrix has 14,502 genes, many correlated. PCA provides a lower-dimensional representation for geometric fitting. Because PCs can reflect technical or tumour-specific effects, we inspect associations with tumour and assay rather than automatically interpreting the leading PCs as EMT."

Q4. Why balance tumours?

"Some tumours contribute many more cells, including a large share of late-EMT cells. Capping the fit at 800 cells per tumour reduces sample-count dominance in PCA fitting. It does not remove confounding or ensure identical biological diversity across tumours."

Q5. What does the permutation test establish?

"It tests whether the observed fit statistic is unusual relative to a null created by independently shuffling the first k–1 PC coordinates. It does not distinguish EMT-driven geometry from tumour- or assay-driven geometry, so the p-value is only one part of the evidence."

Q6. Why not automatically name vertices?

"Vertex names are biological interpretations, not intrinsic outputs of PCHA. We examine state labels, marker genes, tumour composition and robustness first. If the evidence is mixed, we retain neutral labels rather than forcing a biological name."

Q7. What is the biggest limitation?

"Tumour identity and assay type may explain more of the leading expression variation than EMT. We address this with assay-restricted fits and leave-one-tumour-out analyses, but these cannot eliminate every confounder or replace external validation."

Q8. What would convince you that the result is meaningful?

"I would look for reproducible vertex expression programmes, consistent associations with the independently defined EMT labels, representation across tumours, stability under the planned robustness analyses, and ideally confirmation in independent data. A significant fit statistic alone is insufficient."

A good answer includes a limitation

Scientific confidence is not the same as certainty. A precise account of what the method can establish—and what it cannot—is more credible than an overconfident interpretation.

14. Implementation audit: what to verify in the code

The pipeline summary provides a strong overview, but several implementation-specific details cannot be safely inferred from script names alone. Verify these before presenting the work as fully understood.

Item to inspect	What to establish	Why it matters
01_build_anndata.py	Normalisation denominator; gene filtering; mito/ribo gene definition; missing values; gene order.	Defines the actual input matrix.
02_pca.py	Sampling per tumour; centring; number of PCs; score scaling; projection formula; assay diagnostics.	Defines the fitted coordinate space.
run_fits.sh and fit code	Exact ESV definition; objective; delta convention; starts; volume criterion; optimisation settings.	Defines the model and selection evidence.
Permutation code	Statistic; shuffle implementation; whether the observed and null fits use comparable starts; random seeds.	Defines the null and p-value meaning.
04_describe_simplex.py	Weight computation; specialist threshold; marker method; enrichment background; FDR; label joins.	Defines biological summaries.
05_robustness.py	Exact nophase filter; scRNA-only definition; LOTO procedure; what seed changes; vertex matching.	Defines what robustness results test.
PREREGISTERED_PREDICTIONS.md	Date, exact predictions, decision criteria, deviations and outcomes.	Separates a priori expectations from post hoc interpretation.

14.1 A minimum results checklist

- Record the chosen k and the evidence for it; do not infer it from the pipeline plan.
- Report ESV values and permutation results with the exact test statistic and null description.
- Show PC diagnostics for tumour, assay and EMT state.
- Show cell-state and tumour distributions for every interpreted vertex.
- Report how many cells satisfy the specialist rule and how that changes under alternative thresholds if tested.
- Compare all-gene and KS-panel conclusions without calling the panel an independent validation set.
- Report robustness failures as well as successes.
- State which conclusions are exploratory and which were preregistered.

Important

This guide is not a substitute for inspecting the source code and actual result files. In particular, the precise ESV formula, permutation statistic, nophase operation and enrichment implementation must be verified rather than guessed.

15. Glossary, equations and revision questions

Glossary

Term	Plain-language meaning
AnnData	A data structure commonly used to store a gene-expression matrix plus cell and gene metadata.
Archetype	A fitted extreme profile used to represent observations as mixtures.
Assay	The experimental measurement method, here including whole-cell scRNA-seq and multiome nuclei.
Batch effect	Systematic technical variation associated with processing, assay or experimental batch.
Cell state	A description of a cell's molecular or functional programme; may be continuous or overlapping.
Convex combination	A weighted average with non-negative weights that sum to one.
ESV	The fit-related curve used by this pipeline; verify its exact definition in the code.
EMT	Epithelial–mesenchymal transition, a programme of changing epithelial and mesenchymal-associated features.
Gram matrix	A matrix of inner products; here G = XX■ summarises similarities among cells.
LOTO	Leave-one-tumour-out sensitivity analysis.
PCA	Principal Component Analysis; identifies orthogonal directions capturing variance.
PCHA	Principal Convex Hull Analysis, an archetype-fitting method.
PC loading	The weight assigned to a gene in a principal component.
PC score	A cell's coordinate along a principal component.
Simplex	The geometric object defined by archetype vertices: segment, triangle, tetrahedron, and higher-dimensional analogues.
Specialist	A cell whose largest archetype weight exceeds its second-largest by the specified threshold.

Core equations

CP10k(g,c) = count(g,c) / total_counts(c) × 10,000

log1p(x) = log(1 + x)

G = XX■ (cell-by-cell Gram matrix for centred X)

z■ = Σ■ α■■ x■, with α■■ ≥ 0 and Σ■ α■■ = 1

x■ ≈ Σ■ β■■ z■, with β■■ ≥ 0 and Σ■ β■■ = 1

Specialist if w(1) − w(2) ≥ 0.35

Permutation p = (1 + b) / (1 + B), with B = 200

Revision questions

- **Foundations:** Why are raw counts not directly comparable across cells? What does CP10k correct, and what does it not correct?
- **PCA:** What is a PC? Why might PC1 capture assay type rather than EMT?
- **Geometry:** What makes a weight vector a convex combination? Why is an archetype not necessarily an actual cell?
- **Model choice:** What is the difference between an ESV elbow and a permutation test? What null does the permutation test use?
- **Interpretation:** Why are weights not probabilities? What evidence is needed before naming a vertex?

- **Confounding:** Why is tumour balancing not equivalent to batch correction? Why can batch correction remove biology?
- **Robustness:** What threat does LOTO address? Why is it not external validation?
- **Scientific claims:** What result would make you reject or weaken the preferred EMT interpretation?

Final mental model

Preprocess the counts → reduce dimensions with PCA → fit candidate simplices with PCHA → interpret weights against independent EMT labels → challenge the interpretation with tumour, assay and sensitivity analyses. The geometry is a model; the biological story must be earned by evidence.

End of guide

Prepared from the pipeline specification supplied in the conversation. All diagrams and toy values are conceptual; no fitted results from GSE288137 are represented in this document.